

October 2, 2026

The Editor-in-Chief
Computers in Biology and Medicine

Dear Editor,

Please consider the enclosed manuscript, “Leakage-controlled stacking and recursive feature elimination for type 2 diabetes prediction: a rigorous evaluation on the PIMA dataset”, for publication as a Research Article in *Computers in Biology and Medicine*.

Stacking ensembles and wrapper feature selection are widely reported for diabetes prediction, including in this journal, often with very high accuracy on the PIMA Indians Diabetes dataset. Such results are hard to interpret when imputation or feature selection is fitted before validation and when differences between models are not tested statistically. Our study takes a methodological rather than an algorithmic approach:

- every data-dependent step (imputation, feature engineering, scaling, RFECV and stacking) is re-fitted inside each validation split;
- eleven learners are compared with and without RFECV using 5×10 -fold repeated cross-validation, corrected confidence intervals and multiplicity-adjusted tests, and confirmed on an untouched hold-out set with DeLong tests and bootstrap intervals;
- the evaluation goes beyond discrimination to calibration, sensitivity-targeted screening thresholds, decision curves, a missing-data sensitivity analysis and SHAP explanations.

The main finding is deliberately reported without overstatement. RFECV produces compact, clinically plausible feature sets, and the stacking ensemble gives the highest observed hold-out sensitivity and MCC. However, it does not discriminate better than regularised logistic regression once validation is leakage-free, and class weighting causes systematic risk overestimation that accuracy-based reporting would hide. We believe these results, together with fully reproducible code, will be useful to readers developing and reviewing clinical prediction models.

This manuscript is original, has not been published previously and is not under consideration elsewhere. The author has no competing interests to declare. All code and results are publicly

available.

Yours sincerely,

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